COMPLETE LEARNING ROADMAP

AI & Machine Learning

A chapter-wise syllabus from Python foundations to production-ready generative AI

A practical curriculum for building a strong foundation in artificial intelligence and machine learning, progressing from mathematics and data preparation to deep learning, generative AI, responsible practice, and deployment.

33 chapters
Core subjects and applied practice

6 learning parts
Foundations through production systems

Capstone included
Build, evaluate, and present a complete project

01 Foundations

Chapter 1: Introduction to AI and Machine Learning

- AI, machine learning, deep learning, and generative AI
- Narrow and general AI; real-world applications
- Supervised, unsupervised, semi-supervised, and reinforcement learning
- Typical AI project lifecycle

Chapter 2: Python for AI/ML

- Variables, data types, control flow, functions, and classes
- Files, exceptions, modules, and virtual environments
- NumPy arrays and vectorized operations
- pandas, Matplotlib, Seaborn, and notebooks

Chapter 3: Linear Algebra

- Scalars, vectors, matrices, and tensors
- Matrix operations, dot products, norms, and distances
- Linear transformations and systems of equations
- Eigenvalues, eigenvectors, and data representation

Chapter 4: Calculus for Machine Learning

- Functions, limits, and derivatives
- Partial derivatives, gradients, and the chain rule
- Computational graphs and gradient-based optimization
- Backpropagation intuition

Chapter 5: Probability and Statistics

- Descriptive statistics and distributions
- Probability, conditional probability, and Bayes' theorem
- Random variables and common distributions
- Sampling, confidence intervals, hypothesis tests, and correlation

Chapter 6: Data Collection and Preparation

- Data sources, formats, inspection, and cleaning
- Missing values, duplicates, and inconsistent records
- Categorical encoding, scaling, and exploratory analysis
- Data leakage, sampling bias, and data splits

02 Core Machine Learning

Chapter 7: Machine Learning Workflow

- Features, labels, models, and predictions
- Training, validation, testing, and baselines
- Underfitting, overfitting, and generalization
- Reproducible experiments and scikit-learn pipelines

Chapter 8: Regression

- Simple, multiple, and polynomial linear regression
- MAE, MSE, and RMSE loss measures
- Assumptions, residuals, and diagnostics
- Ridge, Lasso, and Elastic Net regularization

Chapter 9: Classification

- Logistic regression and decision boundaries
- Binary and multiclass classification
- Naive Bayes and k-Nearest Neighbors
- Support Vector Machines and class probabilities

Chapter 10: Model Evaluation

- Confusion matrix, accuracy, precision, recall, and F1
- ROC curves and AUC
- Regression metrics and error interpretation
- Class imbalance and problem-appropriate metrics

Chapter 11: Trees and Ensemble Learning

- Decision trees and splitting criteria
- Bagging and Random Forests
- Boosting and gradient boosting
- XGBoost, LightGBM, CatBoost, and feature importance

Chapter 12: Model Selection and Tuning

- Cross-validation and hyperparameter tuning
- Grid search and randomized search
- Feature selection and model comparison
- Learning curves and validation curves

Chapter 13: Unsupervised Learning

- Clustering concepts and distance measures
- k-Means and selecting cluster count
- Hierarchical clustering and DBSCAN
- Cluster evaluation and interpretation

Chapter 14: Dimensionality Reduction

- Purpose and trade-offs of dimensionality reduction
- Principal Component Analysis (PCA)
- t-SNE and UMAP for visualization
- Feature extraction versus feature selection

Chapter 15: Anomaly Detection and Recommenders

- Outlier detection and Isolation Forest
- Density-based anomaly detection
- Collaborative and content-based filtering
- Ranking metrics and cold-start problems

Chapter 16: Time-Series Machine Learning

- Trend, seasonality, and time-series structure
- Time-aware data splitting
- Lag and rolling-window features
- Forecasting metrics and ML forecasting approaches

03 Deep Learning

Chapter 17: Neural Network Fundamentals

- Neurons, layers, weights, and biases
- Activation and loss functions
- Forward propagation, gradient descent, and backpropagation
- Implement models using PyTorch or TensorFlow

Chapter 18: Deep Learning Training Techniques

- SGD, Momentum, Adam, and learning-rate schedules
- Initialization, normalization, and dropout
- Batch size, epochs, and early stopping
- GPU training and experiment tracking

Chapter 19: Computer Vision

- Images as tensors; convolution and pooling
- Convolutional Neural Networks (CNNs)
- Image classification, detection, and segmentation
- Transfer learning for vision tasks

Chapter 20: Natural Language Processing

- Text cleaning, tokenization, and TF-IDF
- Word embeddings and text classification
- Sentiment analysis and sequence modeling
- Language-system evaluation

Chapter 21: Sequence Models

- Recurrent Neural Networks (RNNs)
- Vanishing and exploding gradients
- LSTM and GRU networks
- Sequence-to-sequence applications

Chapter 22: Transformers and Attention

- Attention, self-attention, and positional information
- Transformer encoders and decoders
- Pretraining and fine-tuning
- BERT-style, GPT-style, and non-text applications

Chapter 23: Generative Deep Learning

- Generative versus discriminative models
- Autoencoders and Variational Autoencoders
- Generative Adversarial Networks and diffusion models
- Generation quality and evaluation challenges

04 Modern Generative AI

Chapter 24: Large Language Model Fundamentals

- Tokens, tokenization, and context windows
- Pretraining, inference, and decoding
- Temperature and generation controls
- Hallucinations, limitations, and knowledge boundaries

Chapter 25: Prompting and Structured Outputs

- Instruction design and prompt structure
- Few-shot prompting and examples
- Schema-constrained structured output
- Tool calling, prompt testing, and evaluation

Chapter 26: Retrieval-Augmented Generation

- Embeddings and semantic search
- Document chunking, indexing, and vector databases
- Retrieval, reranking, and grounded generation
- Citations, RAG evaluation, and failure analysis

Chapter 27: Fine-Tuning and Model Adaptation

- Choosing prompting, RAG, or fine-tuning
- Dataset preparation and supervised fine-tuning
- Parameter-efficient tuning such as LoRA
- Validation, overfitting, and model versioning

Chapter 28: AI Agents and Multimodal Systems

- Agent concepts, planning, and orchestration
- Tools, APIs, bounded autonomy, and human approval
- Text, image, audio, and video inputs
- Multimodal evaluation, security, and reliability

05 Responsible AI and Production

Chapter 29: Responsible and Explainable AI

- Fairness, bias, and representation
- Privacy and data protection
- Interpretability and explainability
- Model cards, data documentation, and human oversight

Chapter 30: MLOps and Model Lifecycle

- Data and model versioning
- Experiment tracking and reproducible training
- Model registries and deployment lifecycle
- Monitoring, drift detection, retraining, and rollback

Chapter 31: Deployment and Scaling

- Prediction APIs and batch or online inference
- Model packaging and dependency management
- Containers and cloud deployment concepts
- Latency, throughput, cost, and edge deployment

Chapter 32: AI Security and Reliability

- Input validation and data-quality checks
- Sensitive-data protection and access controls
- Adversarial inputs and model abuse
- Failure testing, logging, and incident response

06 Projects and Specializations

Chapter 33: End-to-End Capstone Project

- Define a problem, target users, and measurable success criteria
- Explore data and establish a baseline
- Train and compare at least two suitable models
- Evaluate on held-out data; analyze errors and limitations
- Deliver a simple deployment or usable demonstration
- Document results, reproducibility, and responsible-AI considerations

Optional Specialization Topics

- Reinforcement learning
- Robotics and autonomous systems
- Speech recognition and audio generation
- Graph machine learning
- Causal inference
- Federated and privacy-preserving learning
- Search, ranking, and recommender systems
- Scientific machine learning

Suggested Practical Tools

Programming: Python

Data work: NumPy, pandas, Matplotlib, Seaborn

Classical ML: scikit-learn

Deep learning: PyTorch or TensorFlow

Generative AI: Hugging Face tools and model APIs

Development: Jupyter, Git, and a code editor

Deployment: FastAPI or Flask, Docker, and cloud services

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